

Brain Wave Pattern Classification: Towards the Design of an Effective Online Classifier

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Abstract

The use of commercial Brain Computer Interface devices has become a trend in different research fields, like Neuroscience and Machine Learning. The purpose of this research is to use a single electrode commercial EEG tool and obtain a good recognition rate by means of feature extraction and basic supervised and unsupervised learning approaches. The data is composed on 35 trials consisting of attention and meditation signals obtained from guiding a virtual character through a specific path. Both methods used a 85-15 basis. The supervised method provided an 80% recognition rate while the unsupervised one provided a 60% recognition rate with an average processing time of around 132 seconds.

Keywords

Brain Computer Interface, Brain-Machine Interface, Support Vector Machine, Self-Organizing Maps, Virtual Reality

1 Commercial Brain Computer Interfaces: A Current Technology Trend

The use of Brain Computer Interfaces for commercial and research purposes is gaining hype and impulse due to its competitive advantage and suitability for the combination of

innovative technologies such as, Human Augmentation, IoT Platforms or Affective Computing [1].

Commercial BCI devices measure signals from the brain and turn them into outputs that provide value to the customer. As with many other novel technologies, it is currently unclear in what situations BCI devices can provide maximum value for the largest number of users. Several manufacturers are currently exploring these questions by offering commercial BCI-like devices. These companies include Open BCI (<http://openbci.com>), Emotiv (<http://www.emotiv.com>) and Neurosky (<http://www.neurosky.com>) among others [2]. The success of widespread dissemination of commercial BCI devices depends on reducing the barriers to acquiring and using these systems. This requirements entails several challenges that relate mainly to cost and ease of use. Reducing cost is the of a traditional BCI system (usually at least 5000 dollars) is mainly a technical problem that can be solved, but does require appropriate resources. One of the approaches included in this research is by means of recognition algorithms of output signals obtained by one of these devices.

1.1 FFT Based BCIs

Basic BCI research and development is based predominantly on recording and analysis of electrophysiological brain signals. The number of channels that are recorded usually varies from 8-64 for EEG. The brain signals recorded from these modalities vary substantially in their amplitudes and frequencies; these waves, preferred range of processing are specified in Table 1[5]:

Table 1: Basic human brain waves and ranges

Name	Frequency range [Hz]	Preferred range [Hz]
Delta [δ]	0.5-2.75	0.5-3.0
Theta [θ]	3.5-6.75	2.0-7.0 or 4.0-8.0
Low Alpha [$l - \alpha$]	7.5-9.25	8.0-10.0
High Alpha [$h - \alpha$]	10.0-11.75	10.0-12.0
Low Beta [$l - \beta$]	13-16.75	SMR 12.0-15.0 Low Beta 15.0-18.0
Low Gamma [$l - \gamma$]	31-39.75	Mid Gamma
Mid Gamma [$m - \gamma$]	41-49.75	High Gamma

The use of Fast Fourier Transform BCI (FFT-BCI) based 3D virtual environments has been implemented effectively in different application fields; it is rather a new technological tool, which may be exploited to enhance motor retraining [3]. A simple development with software like Unity3D (<https://unity3d.com>) can be combined with other assistance technologies [4] or interfaces in order to improve a specific system's performance with an end user's feedback. Recently, the use of this combination is representing an innovative treatment for conditions like ADHD [6]. Usually, FFT-BCI based devices obtain the power specific frequency bands that come from the raw EEG signal sample over a sampling period of time from the active electrodes and perform a ratio calculation that then becomes a user

output. For this research purpose, a TGAM1 ASIC device, like the Neurosky Mindwave [7] with a single electrode tool is used. The output signals from this device include attention and meditation levels (from 0-100%) based on a power band calculation algorithm.

1.2 Brain Wave Signal Classification System

In order to obtain the desired signals to be described in the next chapter, the use of the proper signal obtention system was required. The corresponding system uses the configuration shown in Figure 1. An Arduino Uno board was used as an RF/RS232 data bridge between the headset and the computer script and an Arduino Micro was used for keyboard simulation purposes. The aim of this research is to elaborate a method for recognizing brain wave patterns for authentication or command application development [8].

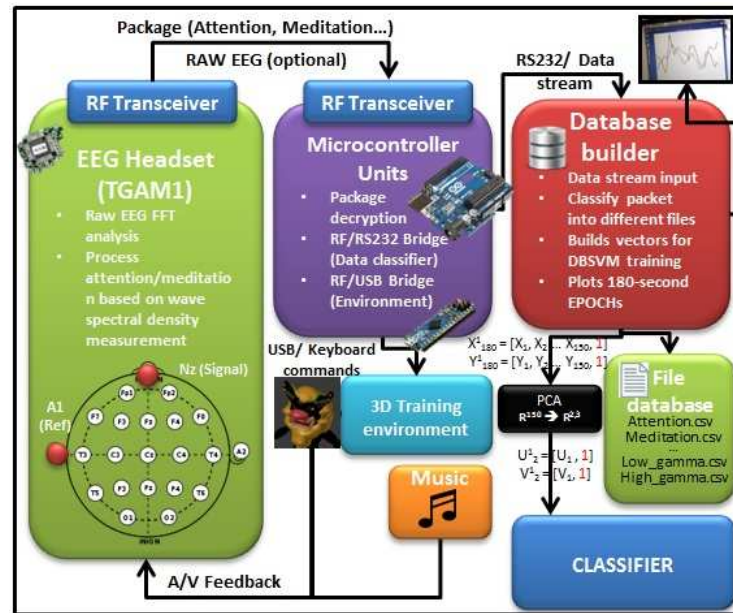


Figure 1: System Description

2 Supervised and Unsupervised Machine Learning Approaches

The purpose of analyzing brain wave patterns is to recognize the use of specific commands and enhance the capabilities of a low cost single electrode device as described before. However, the use of special Machine Learning techniques require the extraction of special features according to the common BCI signal extraction methods [9]. From previous experiments [8], it was concluded that a series of alternating attention/meditation signals over time for a 1 Hz frequency device such as the Neurosky MindWave could be generated on a

specific period of time.

However, sometimes these signals might not achieve the expected peaks or synchronized when performing online. In order to enhance the comprehension of the obtained signals, the use of Machine Learning techniques could be fitted in order to understand the purpose of a especial signal over time based on the characteristics from the obtained samples. In this research, so far two approaches of supervised (Density Based Support Vector Machine) and unsupervised (Kohonen layer Self-Organizing Map) learning methods were used to compare their feasibility and their application for embedded system execution purposes. These methods are described below.

2.1 Density Based Support Vector Machines

Machine Learning techniques are generally helpful in feature extraction among EEG signals, since the recorded signals usually tend to have variable bias and uncertain changes, which make the feature classification difficult. The Support Vector Machine (SVM) model proposed by Dr. Vapnik has been used as a form of supervised learning models (SLM) in detecting certain features in the captured EEG data and in many other systems [10].

Unfortunately, the sensitivity of SVM to outliers is a weak point for this algorithm. One of the approaches for reducing this sensitivity is the Density Based Support Vector Machine (DBSVM) method, which is based in the simple expression of population density according to Equation 1:

$$Populationdensity = \frac{numberofpopulation}{area} \quad (1)$$

Although the concept of population density is used to develop the DBSVM model, the formula is actually based on two separate algorithms to remove the outliers based on a vectorfs total Euclidean or Mahalanobis distance between its averages [11].

2.1.1 DBSVM with Euclidean distance

For instance, letfs suppose a set of n numbers of two-dimensional data $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is given. The Euclidean distance between two points is given by Equation 2:

$$D(1, 2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \quad (2)$$

The next step is to sum up all the distances for each point (i.e. $d_1 = [D(1, 2) + D(1, 3) + \dots + D(1, n)]$) where n is the number of data points in the data set. The total distance for all data points of one data set is needed to calculate the average distance that will be used to determine data points inside and outside of densely populated area. Equation 3 shows the expression to calculate this:

$$Average_d = \frac{\sum_{j=1}^N \sum_{i=1}^N \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}}{n}; \text{ if } d_i > Average_d \rightarrow x_i = outlier \quad (3)$$

After calculating $Average_d$ by formula, those points that are not considered as outliers will become the new training data set in which the points will be considered important data points.

2.1.2 DBSVM with Mahalanobis Distance

The Mahalanobis distance is the distance between x from quantity μ . This distance is based on the correlation between variables or the variance-covariance matrix. To determine the distance from two data points, the formula from Equation 4 is used:

$$D_m(x) = \sqrt{(x - y)^T S^{-1} (x - y)} \quad (4)$$

Therefore, as used for the Euclidean Distance average calculation, the average Mahalanobis distance for excluding outliers is as described in Equation 5:

$$Average_d = \frac{\sum_{i=1}^N \sqrt{(x_i - y)^T S^{-1} (x_i - y)}}{n} \quad (5)$$

2.2 Self-Organizing Maps (SOM)

The Kohonen Layer Self-Organizing Maps provide a way of representing multidimensional data in much lower dimensional spaces, usually one or two dimensions (vector quantization). In its simplest form it produces a similarity graph of input data. It converts nonlinear statistical relationships between high-dimensional data into simple geometric relationships of their image points on a low-dimensional display. Usually, a regular two-dimensional grid of nodes is used. Assume vector $\mathbf{x} = \{\xi_1, \xi_2, \dots, \xi_n\}^T \in \mathbb{R}^n$. With each element in the SOM array, we associate a parametric real vector $\mathbf{m}_i = \{\mu_1, \mu_2, \dots, \mu_n\}^T \in \mathbb{R}^n$ that is called the model. Assuming a general distance measure between $\mathbf{x}$ and $\mathbf{m}_i$ denoted $d(\mathbf{x}, \mathbf{m}_i)$, the image of an input vector $\mathbf{x}$ on the SOM array is defined as the array element m_c that matches best with $\mathbf{x}$. In this research, the most common approach for determining the winner c , described by Equation 6 is used.

$$c = \operatorname{argmin}_i \{d(\mathbf{x}, \mathbf{m}_i)\} \quad (6)$$

Differing from the traditional vector quantization, $\mathbf{m}_i$ is to be defined in such a way that the mapping is ordered and descriptive of the distribution of $\mathbf{x}$. For effective mapping, it will suffice that the distance measure is defined over all occurring $\mathbf{x}$ items and a sufficiently large set of models $\mathbf{m}_i$ [12].

3 Signal Acquisition, Conditioning and Feature Extraction

3.1 The Virtual Training Environment

The proposed training environment was designed with the use of Blender for the animated models and structures and Unity3D for its ease of use and deployment. The environment has a series of features such as music and interaction with diverse experiment variables. The environment setup was created for the test subjects to familiarize with the objects and understand the way of interacting with them; an additional relaxation music track can be used for aiming in obtaining higher meditation behaviors. The purpose of the training task is to guide the player (training subject) throughout the predefined path, as specified in Figure 2.

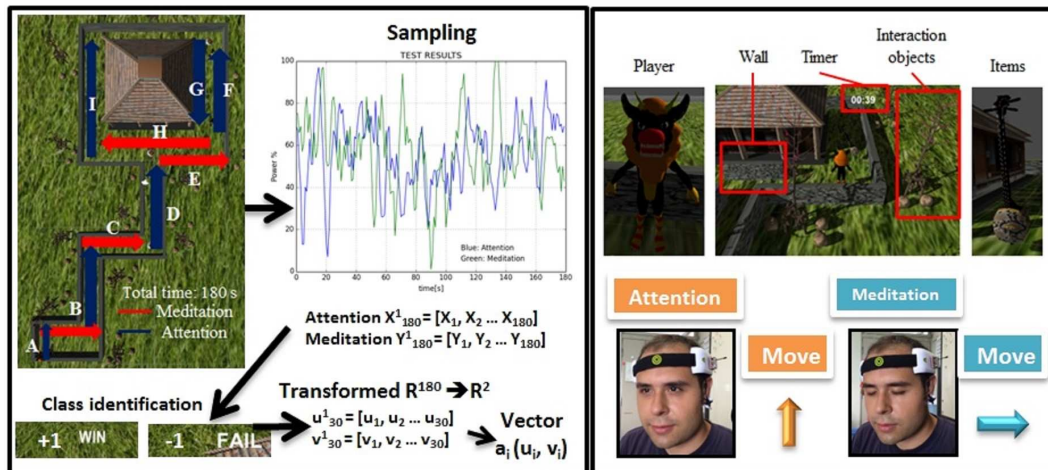


Figure 2: The Virtual Training Environment

When concentrated, the user can move the player forward and then move the player to the right if the user begins to enter meditation. In either case, the user can stop the player by blinking the eyes constantly until the player stops due to the artifact noise produced across the path, there are four main items that the subject requires to obtain. If the series of mind-waves came into the expected pattern in the required time, the attempt will be classified as successful and the class +1 will be labeled for the output 180 second EPOCH. However, if the test subject is not able to perform the entire track along the timeframe, an unsuccessful attempt is recorded with the -1 label. The resulting vectors will be used as samples for later mapping in the proposed classifiers. Brain waves are dependent on each test subject, and so the performed wave patterns and headset calibration. Due to this factor, the training requires to be performed individually. Fortunately, the device has a fast self-calibration routine and a trained subject can easily adapt in controlling the headset in minutes. In order to have a good classifier, 35 samples were obtained from the test

subject. The testing area was a quiet room with the test subject sitting in front of a screen displaying the 3D environment and the data processing script running in the background for obtaining the corresponding data sets. Whenever the test subject was ready to perform a test a red button was pressed to restart the environment and script.

3.2 Feature Extraction and Dimensionality Reduction

Each sample has a length of 180 samples and consists of both an Attention **A** and Meditation **B** vectors. In order to obtain a feature vector that could provide a good real time classifier by using DBSVM, different test vectors were utilized. However, the array that could provide the best classification results was formed by use of empirical Equation 7 [13] for feature vector **X**:

$$X = \frac{\phi A^2(A - B)}{A + B + \phi}; \phi = 0.001 \quad (7)$$

Once the matrices have been associated with the feature, the PCA matrix *P* is obtained for matrix **X** [14]. The two main components (PCA2) of the transform matrix *Y* detailed by Equation 8 were used to map the outputs in the two-dimensional space for the DBSVM approach.

$$XP = Y \quad (8)$$

For the unsupervised learning approach, whitening was applied to the feature vectors for different cases, three (PCA3), five (PCA5), seven (PCA7) and ten (PCA10) dimensions. The eigenvalue plots after applying Singular Value Decomposition [15] for the Attention **A**, Meditation **B** and Feature vector **X** suggest that at least ten dimensions can be considered as the most important number of components to perform training for the Self-Organizing Map implementation.

4 Results

4.1 DBSVM Implementation

After performing whitening (PCA2) to feature vector **X**, outlier removal and training was performed for the obtained samples on an 85-15 basis (85% training, 15% test). The kernels that were used for the SVM implementation were the linear, RBF and polynomial degree 3. The LinearSVC kernel uses a different code library that allows a more flexible modeling and better choice of penalties and loss function scaling [16]. The classification success rate for the test data set could be augmented if a soft margin is implemented. The resulting classifiers are presented on Figure 3 and the resulting classification rates are shown on Table 2:

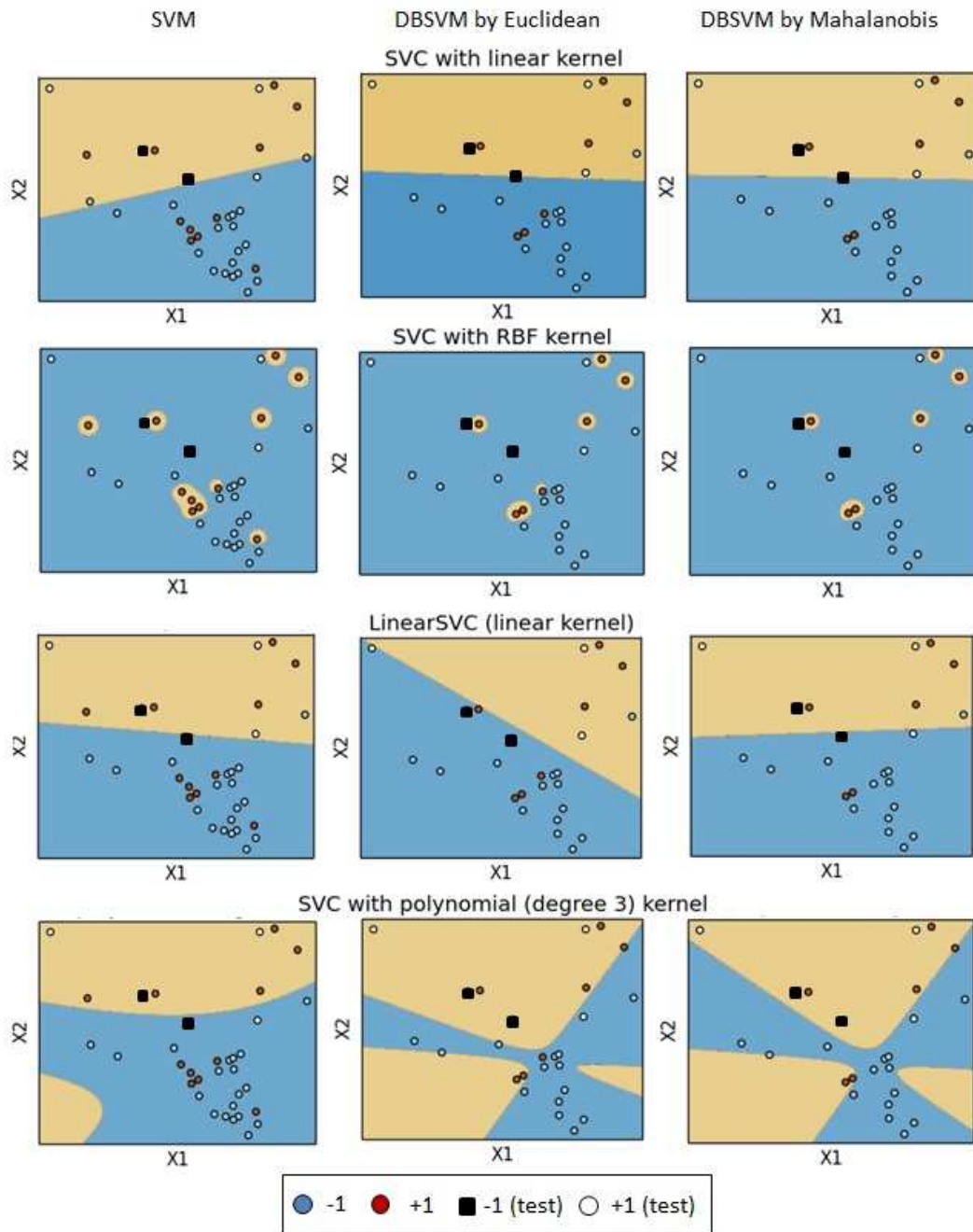


Figure 3: SVM and DBSVM classifiers

Table 2: SVM and DBSVM classification results

Kernel	Normal	Euclidean	Mahalanobis
Linear	20%	60%	60%
RBF	40%	40%	40%
LinearSVC	80%	80%	60%
Polynomial	60%	20%	20%

4.2 SOM Implementation

In order to test the capabilities of Unsupervised Learning Method for classification, a small classification test was performed on the feature vector with a dimension reduction of three by PCA [17], because it allows distinguishing the colors of each of the regions.

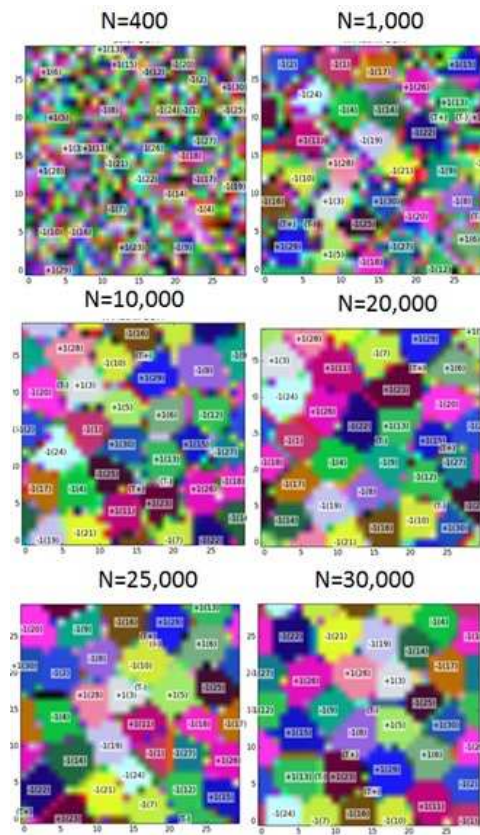


Figure 4: SOM PCA3 at different iterations

A 30x30 grid Kohonen layer map with Gaussian kernel and exponential learning rate was implemented for the distance function described previously. At a glance, it looks that the map is not performing well when the number of iteration is low ($n < 10000$). However, once the number of iteration has been increased ($n > 20000$), the areas are getting defined and the map seems to be formed clearly, as presented in Figure 4 (the training was independent for each picture). The use of different PCA components for the main Feature vector X and other features composed by the Attention and Meditation values were tested with the Python library [18]. The average processing speed with an Intel Core i7 processor at 2.30 GHz computer for each classification trial was performed for five iterations of the SOM on each trial. 10-fold cross validation tests were performed in order to provide the average recognition rate for the test vectors. Table 3 presents these results for different number of iterations (I = 20,000, II=25,000 and III=30,000). The performance rate is the ratio between the recognition rate and the average SOM processing time.

Table 3: 10-fold cross validation results for different features and recognition rates by SOM

Feature	PCA3			PCA5			PCA7			PCA10		
A	I	II	III	I	II	III	I	II	III	I	II	III
Recognition rate	0.52	0.52	0.68	0.4	0.4	0.52	0.6	0.64	0.52	0.4	0.56	0.40
Processing time (s)	455.95	572.90	686.31	528.89	645.54	786.77	568.69	737.39	855.77	670.03	828.05	982.87
Average time(s)	91.19	114.58	137.262	105.77	129.10	157.35	113.73	147.47	171.15	134.00	165.60	196.57
Performance ratio	0.57	0.45	0.49	0.37	0.30	0.33	0.52	0.43	0.301	0.30	0.34	0.20
B	I	II	III	I	II	III	I	II	III	I	II	III
Recognition rate	0.20	0.24	0.32	0.20	0.20	0.60	0.24	0.52	0.40	0.48	0.40	0.76
Processing time (s)	474.03	595.53	709.85	567.26	688.07	837.12	594.92	745.91	882.59	662.13	833.47	966.87
Average time(s)	94.81	119.11	141.97	113.452	137.614	167.42	118.98	149.18	176.52	132.43	166.69	193.374
Performance ratio	0.21	0.20	0.23	0.18	0.15	0.36	0.20	0.35	0.23	0.36	0.24	0.39
X	I	II	III	I	II	III	I	II	III	I	II	III
Recognition rate	0.26	0.32	0.4	0.4	0.44	0.56	0.40	0.60	0.64	0.60	0.60	0.60
Processing time (s)	444.21	557.27	665.48	518.43	698.27	839.45	608.77	744.97	865.29	658.90	797.70	1027.56
Average time(s)	88.84	111.45	133.10	103.69	139.65	167.89	121.75	148.99	173.06	131.78	159.54	205.512
Performance ratio	0.29	0.29	0.30	0.39	0.32	0.33	0.33	0.40	0.37	0.46	0.37	0.29
A+B	I	II	III	I	II	III	I	II	III	I	II	III
Recognition rate	0.44	0.56	0.60	0.40	0.36	0.64	0.20	0.24	0.44	0.40	0.12	0.32
Processing time (s)	484.21	1367.57	777.77	518.97	643.56	777.77	575.89	711.74	2006	640.12	811.07	1023.04
Average time(s)	96.84	273.51	155.55	103.79	128.71	155.55	115.18	142.34	401.2	128.02	162.21	204.61
Performance ratio	0.45	0.20	0.39	0.39	0.28	0.41	0.17	0.17	0.11	0.31	0.74 E-2	0.16
A-B	I	II	III	I	II	III	I	II	III	I	II	III
Recognition rate	0.44	0.4	0.56	0.48	0.52	0.4	0.48	0.44	0.44	0.4	0.56	0.60
Processing time (s)	483.21	643.56	777.77	518.97	643.56	777.77	518.97	643.56	777.77	658.90	797.70	1027.56
Average time(s)	96.64	128.71	155.55	103.79	128.71	155.55	103.79	128.71	155.55	131.78	159.54	205.51
Performance ratio	0.45	0.31	0.36	0.46	0.40	0.26	0.46	0.34	0.28	0.30	0.35	0.29
A/B	I	II	III	I	II	III	I	II	III	I	II	III
Recognition rate	0.28	0.54	0.54	0.20	0.20	0.32	0.12	0.36	0.36	0.40	0.64	0.72
Processing time (s)	448.45	570.56	679.58	518.97	643.56	777.77	518.97	643.56	777.77	662.13	833.47	966.87
Average time(s)	89.69	114.11	135.916	103.79	128.71	155.554	103.79	128.71	155.55	132.42	166.69	193.37
Performance ratio	0.31	0.47	0.40	0.19	0.15	0.21	0.12	0.28	0.23	0.30	0.38	0.37

5 Conclusions

From this research, we have concluded from the results that Feature vector X provided the best classification/processing time ratio for the maximum number of components available (PCA10) for the attention/meditation brainwave patterns for the Unsupervised approach. However, the fact that the recognition rate is not very high suggests that a different feature vector should be found for this methodology. For this particular problem, the selected fea-

ture was the one that worked best after trying various models with the use of PCA after SVD. There can be other mapping approaches that can be performed based on the Attention/Meditation signal correlation. As for the DBSVM classifier, the use of a more flexible linear kernel for the training could obtain a higher learning rate without soft margin. If soft margin was included, the learning success would be 100%. Nevertheless, the number of test vectors should be increased for a better test accuracy. The performed tests demonstrate the possibility of enhancing the use of low-cost, single electrode commercial FFT-based EEG headsets for simple classification tasks with the aid of a simple training environment with optional music feedback. These methodologies can be ported to an embedded system for the current dataset and verify their use when attempting new trials from the same or different end users.

Further Work

At this current point, the selection of other classification algorithms are taken into account in order to compare the effectiveness of their application to both methodologies from the algorithm perspective. In order to increase the processing time when porting this classifiers to an embedded version, a parallel-algorithm approach (Parallel-SOM and Parallel SVM implementations) will be considered by comparing their efficiency when working with high speed GPU systems with online data acquisition and high resolution graphic display.

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